

MUHAMMAD HAMAD

AI / Computer Vision Developer | LangChain & AI Developer | Flutter Developer

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PROFESSIONAL SUMMARY

Computer Science graduate specializing in AI, Computer Vision, and LangChain-based applications, with additional experience in Flutter mobile development. Skilled in building real-time computer vision systems using OpenCV and MediaPipe, deep learning pipelines with YOLO, and LLM-powered chatbots using LangChain and RAG. Passionate about turning practical ideas into working, portfolio-ready AI applications.

TECHNICAL SKILLS

AI / ML & CV	OpenCV, MediaPipe, YOLOv8, YOLOv9, TensorFlow, Scikit-learn
LLM & AI Tools	LangChain, Grok API, RAG Pipeline, Vector Stores, Embeddings
Programming Languages	Python, Dart, C++
Mobile Development	Flutter, Dart, Android SDK
Backend / Database	REST APIs
Dev Tools	Git, GitHub, VS Code, Android Studio, Google Colab
Soft Skills	Problem Solving, Team Collaboration, Communication

PROJECTS

Final Year Project (FYP) — Classroom Activity Detection System

2025 – 2026

- Developed a real-time classroom monitoring system using computer vision to detect student activities including talking, sleeping, mobile phone usage, standing, eating, hand raising, and fighting.
- Collected and annotated a custom dataset of university-level classroom activities for model training.
- Trained and evaluated two deep learning models: YOLOv8 and YOLOv9, comparing performance on activity detection tasks.
- Implemented live camera feed integration to detect and classify multiple activities simultaneously in real time.

InvisiCloak — Real-Time Invisibility Cloak

2026

- Built a computer vision application that makes a colored cloak appear invisible in real time using color detection and background substitution with OpenCV.
- Implemented HSV color-space masking and background frame capture to seamlessly blend the cloaked region with a pre-captured background.
- Available on GitHub: github.com/TechBros6/Invisi_Cloak

GestureCanvas — AI Virtual Painter

2026

- Developed a hand-gesture-controlled drawing application using MediaPipe hand tracking and OpenCV, allowing users to paint on screen using only finger movements.
- Implemented real-time finger-state detection to switch between drawing mode and a color/eraser selection toolbar.
- Available on GitHub: github.com/TechBros6/GestureCanvas

Framelt — Gesture-Controlled Photo Puzzle

2026

- Built a computer vision game combining two-hand gesture tracking with an interactive jigsaw puzzle: users frame a photo with their fingers, pinch to capture it, and solve the shuffled puzzle.
- Implemented normalized pinch-gesture detection and a state-machine architecture (framing, capture, puzzle-solving) using MediaPipe and OpenCV.
- Available on GitHub: github.com/TechBros6/Framelt

KO Trainer — AI Boxing Game

2026

- Created a pose-estimation-based boxing game where players throw real punches at the webcam to damage an AI opponent's health bar.

- Used MediaPipe Pose to compute elbow-joint angles and detect punches from rapid bent-to-extended arm transitions, with a stateful combat/scoring system.
- Available on GitHub: github.com/TechBros6/KO-Trainer-

AI Chatbot — LangChain + Grok API2026

- Built a conversational AI chatbot using the LangChain framework and Grok API for natural language understanding and generation.
- Designed multi-turn conversation flows with context management and prompt engineering for accurate responses.

YouTube Video Chatbot — RAG Pipeline2026

- Developed an AI-powered chatbot that allows users to add any YouTube video link and ask questions based on the video content.
- Pipeline: Video URL input → transcript loading → text chunking → vector store creation → embedding generation → RAG chain setup → final chain build for Q&A.
- Integrated Grok API as the language model backbone and implemented vector similarity search for accurate context retrieval.

Internship Project — Habit App (Flutter, Frontend)2025

- Designed and developed the complete frontend UI/UX of a habit-tracking mobile application using Flutter and Dart.
- Implemented intuitive navigation, state management, and responsive UI components following Material Design principles.
- Collaborated with the backend team to integrate APIs and ensure seamless data flow across the application.

WORK EXPERIENCE

Flutter Developer Intern | NeusoftixJun 2024 – Sep 2024 • 3 Months

- Developed and maintained the frontend UI of the Habit App using Flutter and Dart during a 3-month internship at Neusoftix.
- Implemented responsive UI components, navigation flows, and state management following Material Design standards.
- Collaborated with the backend team to integrate REST APIs and ensure smooth data handling across the application.
- Gained hands-on experience in professional agile workflows, code reviews, and industry-standard development practices.

EDUCATION

Bachelor of Science — Software Engineering | Lahore Garrison University2022 – 2026

Relevant coursework: Machine Learning, Computer Vision, Mobile App Development, Data Structures, Software Engineering, Database Systems.

CERTIFICATIONS

- Flutter intern certificate

LANGUAGES

English	Professional Proficiency
Urdu	Native